

# AI Research Assistant System

## SYNOPSIS

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# Project Name

AI Research Assistant System

## Description

AI Research Assistant System is a modern, user-friendly web-based platform designed to help students and researchers understand, search, and analyze research documents using Artificial Intelligence. The system allows users to upload PDF documents, process their content, ask questions about the uploaded material, and receive AI-generated answers based on relevant information retrieved from those documents.

The system uses Retrieval-Augmented Generation (RAG) to retrieve relevant document content before generating an answer. This approach helps the assistant provide document-focused responses and makes it easier for users to find useful information from large research papers and study materials.

## Platform / Technology Stack

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| Frontend          | React.js, JavaScript, Tailwind CSS, React Router                 |
| Backend           | Python, FastAPI                                                  |
| AI / RAG          | Large Language Model, Embeddings, Retrieval-Augmented Generation |
| PDF Processing    | PyMuPDF / PyPDF                                                  |
| Vector Storage    | ChromaDB / FAISS                                                 |
| Database          | MongoDB                                                          |
| Authentication    | JWT-based authentication and password hashing                    |
| Development Tools | VS Code, Git, GitHub, Postman                                    |

## Key Features

- User Registration and Login – Secure account creation and authentication.
- PDF Upload – Upload research papers, notes, and other PDF study materials.
- Document Management – View, manage, search, and delete uploaded documents.
- AI Document Analysis – Process and analyze uploaded documents using AI techniques.
- Ask AI – Ask questions about uploaded research documents and receive relevant answers.
- RAG-based Question Answering – Retrieve relevant document content before generating responses.
- Document Summarization – Generate concise summaries of research documents.
- Source References – Display the document and relevant source information used for an answer.
- Chat History – Store and review previous questions and AI responses.
- Responsive Interface – Provide a usable interface across desktop, tablet, and mobile devices.

## System Requirements

**Hardware:** Minimum 8 GB RAM, Intel i3 or equivalent processor, 10 GB available storage, and a stable internet connection.

**Software:** Windows/Linux/macOS, Node.js, Python, MongoDB, VS Code, Git, and required frontend/backend libraries.

**Optional:** A GPU may be useful when running local AI models, but it is not compulsory when using an external AI API.

## Functional Requirements

- Users shall be able to register and log in securely.
- Users shall be able to upload PDF research documents.
- The system shall extract text from uploaded documents.
- The system shall divide document text into smaller searchable chunks.
- The system shall generate embeddings and store them in a vector database.
- Users shall be able to ask questions related to uploaded documents.
- The system shall retrieve relevant document information for each question.
- The AI model shall generate an answer using the retrieved context.
- The system shall maintain chat history and document metadata.
- Users shall be able to manage and delete their documents.

## Non-Functional Requirements

- **Performance:** The system should provide responses within a reasonable time.
- **Security:** User accounts, documents, passwords, and API credentials must be protected.
- **Usability:** The interface should be simple and easy to use.
- **Scalability:** The architecture should support future growth in users and documents.
- **Reliability:** The system should handle invalid files and service errors gracefully.
- **Responsiveness:** The application should work on desktop, tablet, and mobile devices.

## Uses / Applications

- Helps students understand research papers and study materials.
- Helps researchers quickly locate relevant information in large documents.
- Supports document-based question answering and summarization.
- Reduces the time required for manual document searching.
- Can be used as an AI-powered academic research companion.

## Conclusion

AI Research Assistant System aims to be more than a basic chatbot. It is a document-focused research platform that combines React, FastAPI, PDF processing, Retrieval-Augmented Generation, vector search, MongoDB, and Large Language Models. By retrieving relevant information from uploaded documents before generating answers, the system can help students and researchers understand large amounts of study material more efficiently. The project can also be extended in the future with additional document formats, OCR, multilingual support, voice interaction, citation generation, document comparison, and advanced research tools.